Problem

Problem · Coefficients and roots of a cubic polynomial. Let $P(z) = z^3 + az^2 + bz + c$ be a polynomial with complex coefficients, with three roots z_1, z_2, z_3 (not necessarily distinct), so that $P(z) = (z - z_1)(z - z_2)(z - z_3)$ for every complex number z.

1. By expanding, prove that
$$z_1 + z_2 + z_3 = -a, \qquad z_1z_2 + z_1z_3 + z_2z_3 = b, \qquad z_1z_2z_3 = -c.$$
2. Check these relations on the polynomial of the activity, $z^3 - 2z^2 + 4z - 8$.
3. Prove that $z_1^2 + z_2^2 + z_3^2 = a^2 - 2b$.
4. Consider $R(z) = z^3 - 2z^2 + 3z - 5$, which we take for granted can be written $(z - z_1)(z - z_2)(z - z_3)$. Compute $z_1^2 + z_2^2 + z_3^2$ without solving the equation.
5. Deduce that the roots of R are not all real.

Beyond the syllabus · The d'Alembert-Gauss theorem. Every polynomial of degree $n \geq 1$ with complex coefficients factorises in $\mathbb{C}$ as a product of n factors of degree 1. This theorem is not part of the syllabus: it is admitted in question 4 and is not assessed.